

Fractional Follies

Fall 2020 ARML Power Contest

In this exploration, a *program* will be a list of positive fractions in lowest terms. A program can be *applied* to a starting number, which is a positive integer, resulting in another positive integer $P(n)$. To apply program $P = (f_1, f_2, \dots, f_k)$ to the integer n , determine the first fraction in the list so that $f_i n$ is an integer, and that integer is the result of the application. If none of the fractional multiplications produces an integer, then the result of the application is just the original number n . For example, let

$$P = \left(\frac{2}{5}, \frac{15}{7}, \frac{3}{4} \right).$$

The result of applying P to 14 is 30, because $\frac{2}{5} \cdot 14$ is not an integer, but $\frac{15}{7} \cdot 14 = 30$ is an integer. The result of applying P to 9 is 9, because none of the fraction in P yields an integer on multiplication by 9.

Given a program P and a starting integer $n_0 = n$, generate a sequence by recursively applying P to n_i . That is,

$$n_{i+1} = \begin{cases} n_0 & i = 0 \\ P(n_i) & i > 0 \end{cases}.$$

Call this the *sequence generated by P starting from n* . For the P above, the sequence generated by P starting from 14 is thus 14, 30, 12, 9, 9, 9, $\dots$. To avoid repetition, declare a sequence that repeats the same number forever (because no fractions of the program multiply the number to give a different integer) to terminate. So instead of repeating 9 forever, this sequence will just stop the first time 9 is reached: 14, 30, 12, 9.

Problem 1. For each program P and starting number n_0 below, determine the next eight terms of the sequence (that is, terms n_1 through n_8).

- [2] (a) $P = \left(\frac{4}{3}, \frac{3}{2} \right)$, $n_0 = 2$.
- [2] (b) $P = \left(\frac{3}{11}, \frac{847}{45}, \frac{143}{6}, \frac{7}{3}, \frac{10}{91}, \frac{3}{7}, \frac{36}{325}, \frac{1}{2}, \frac{36}{5} \right)$, $n_0 = 10$.

- [2] **Problem 2.** Let $P = \left(\frac{2}{3} \right)$ and let $n_0 = 2^a 3^b$ where a and b are any nonnegative integers. Determine the number at which the sequence generated by P starting from n_0 terminates.

- [2] **Problem 3.** Let $P = \left(\frac{2}{15} \right)$ and let $n_0 = 3^a 5^b$. Determine the number at which the sequence generated by P starting at n_0 terminates.

- [2] **Problem 4.** Let $c = \max\{a, b\}$. Create a program that generates a sequence beginning with $3^a 5^b$ and terminates with 2^c .

Given integers a and b that share no common factors, the program $\left(\frac{b}{a} \right)$ applied to a results in b , and then terminates because b^2/a is not an integer. But if a and b do have

common factors it is not so easy to find a program that generates a sequence starting with a and terminating with b . For example, Let $a = 4$ and $b = 6$. Since $\frac{6}{4}$ is not in lowest terms it cannot be part of a program. The program $P = (\frac{3}{2})$ applied to 4 yields 6—but the sequence does not terminate there! P can be applied to 6 to yield 9. Is there a program that produces a sequence that starts at 4 and terminates at 6? Yes there is: try $P = (\frac{17}{4}, \frac{6}{17})$. (The only thing special about 17 here is that it shares no common factors with 4 or 6.) A similar trick will work for any choice of a and b as long as b is not a multiple of a .

- [2] **Problem 5.** Prove that if $b = ka$ where $k \geq 2$ is an integer, then there is no program P that will generate a sequence starting with a and terminating at b .

These ideas can be extended to create programs that, when started with different specified starting numbers, generate sequences that terminate at specified ending numbers, as long as none of the ending numbers are multiples of any of the starting numbers.

- [2] **Problem 6.** Create a program that, when started with 6 will generate a sequence ending with 2, when started at 8 will generate a sequence terminating at 4, and when started with 9 will generate a sequence terminating at 3. (It can have any behavior for any other starting values.)

As demonstrated in problems 2–4, it is better to think of a program as being applied to the prime power factorization of n_0 and examining what happens to the exponents. That is, given $n_0 = 2^a 3^b 5^c 7^d \dots$ what does the program do with the exponents $a, b, c, d, \dots$?

For any remaining problems, if you are asked to create a program that generates particular sequences given specified starting values, there is no restriction to the behavior of the program if the input is not among the specified values.

- [2] **Problem 7.** Let $P = (\frac{1}{6})$. Prove that if $n_0 = 2^a 3^b$ then the sequence generated terminates at 2^{a-b} if $a > b$, at 3^{b-a} if $a < b$, and at 1 if $a = b$.

- [2] **Problem 8.** Create a program that, if started with 3^b , generates a sequence ending with 1 if b is even and 2 if b is odd.

- [2] **Problem 9.** Let $P = (\frac{7}{15}, \frac{5}{7}, \frac{5}{3}, \frac{2}{5})$. Show that if this program is applied to $n_0 = 3^b$ the sequence will terminate at 1 if $b = 0$ and at 2 if $b > 0$.

- [4] **Problem 10.** Create a program that, if started with $3^b 5^c$, generates a sequence that terminates at 1 if $b = c$, at 2 if $b > c$, and at 4 if $c > b$. Explain why your program performs as claimed.

These programs can get pretty sophisticated! Consider the program

$$P = \left(\frac{5 \cdot 7 \cdot 13}{3 \cdot 11}, \frac{11}{13}, \frac{1}{11}, \frac{3}{7}, \frac{11}{2}, \frac{1}{3} \right).$$

This program, when started with $2^a 3^b$ terminates with 5^{ab} . Yes, this program *multiplies* two

numbers!

Problem 11. Prove this:

- [2] (a) First, show that if $a = 0$ the program will terminate at $5^0 = 1$.
- [2] (b) Otherwise, $n_1 = 2^{a-1}3^b11$. In the next phase of the sequence, only the first three fractions of the program are used. In terms of a and b , what is the value of the term of the sequence just before any fraction other than the first three is about to be used?
- [1] (c) In the next phase, only the fourth fraction is used. What is the value of the sequence after this phase is complete?
- [3] (d) Complete the proof. Make sure to explain the purpose of the $\frac{1}{3}$ at the end of the program.

- [8] **Problem 12.** Let $P = (\frac{3 \cdot 11}{2^2 \cdot 5}, \frac{5}{11}, \frac{13}{2 \cdot 5}, \frac{1}{5}, \frac{2}{3}, \frac{2 \cdot 5}{7}, \frac{7}{2})$. Prove that when started with 2^a , the sequence generated by this program terminates with 13^h , where h is the *Hamming weight* of a —the number of ones in its binary expansion.

The program in problem 1(b) is actually a prime generator. After the initial $n_0 = 10$ the only perfect powers of 10 that appear in the sequence are 10^p where the primes p occur in increasing order. It takes a while, though. While $n_{10} = 10^2$ occurs fairly early, the next power of 10 that occurs is $n_{46} = 10^3$ and then $n_{196} = 10^5$, and 10^7 doesn't show up until n_{500} . Not exactly the most efficient way to find primes!